The Eigenstructure of the Black–Scholes Second-Order Greek Matrix

A note on (S, σ) curvature, its signature, and the at-the-money eigenframe

2026-07-24

Abstract

The Hessian of the Black–Scholes price in spot and implied volatility has signature governed by $\operatorname{sign} d_2$ alone; at the money it factors into a scalar times a matrix of determinant $-\frac{1}{2}$, so its eigenframe is fixed and vanna acts only as a rotation by half the total variance. Eigenvectors are coordinate-dependent; the inertia is not.

1 Setup

Work in the Black–Scholes frame with $r = q = 0$, so $F = S$; fix strike K, maturity T, implied volatility σ, and write $m = \ln(K/S)$. Then

$$d_{1,2} = -\frac{m}{\sigma\sqrt{T}} \pm \frac{\sigma\sqrt{T}}{2}, \qquad d_1 - d_2 = \sigma\sqrt{T}, \qquad d_1 d_2 = \tfrac{1}{4}\sigma^2 T - \tfrac{m^2}{\sigma^2 T},$$

and $\varphi = \varphi(d_1)$ is the standard normal density at d_1. The second-order sensitivities in (S, σ) assemble into the symmetric Hessian

$$H = \begin{pmatrix} \Gamma & \text{Vanna} \\ \text{Vanna} & \text{Volga} \end{pmatrix}, \qquad \Gamma = \frac{\varphi}{S\sigma\sqrt{T}}, \quad \text{Vanna} = -\frac{\varphi\, d_2}{\sigma}, \quad \text{Volga} = \frac{S\sqrt{T}\,\varphi\, d_1 d_2}{\sigma}.$$

H is the curvature of the price as a function of the two state variables a desk marks against; its eigenstructure is the local principal-axis decomposition of that curvature.

2 Signature

Proposition 1. $\det H = \dfrac{\varphi^2\sqrt{T}}{\sigma}\, d_2$, *hence* $\operatorname{sign}\det H = \operatorname{sign} d_2$.

Proof. Directly,

$$\det H = \Gamma\,\text{Volga} - \text{Vanna}^2 = \frac{\varphi^2 d_1 d_2}{\sigma^2} - \frac{\varphi^2 d_2^2}{\sigma^2} = \frac{\varphi^2}{\sigma^2}\, d_2\,(d_1 - d_2) = \frac{\varphi^2}{\sigma^2}\, d_2\,\sigma\sqrt{T} = \frac{\varphi^2\sqrt{T}}{\sigma}\, d_2. \qquad \square$$

Since $\Gamma > 0$ always, the signature is read off d_2: H is positive definite when $d_2 > 0$, rank one when $d_2 = 0$, and a saddle when $d_2 < 0$.

Corollary 1. $d_2 > 0 \iff m < \frac{1}{2}\sigma^2 T \iff S > K\, e^{\sigma^2 T/2}$. *Curvature in (S, σ) is a genuine minimum only sufficiently far in the money; at spot–ATM ($m = 0$) one has $d_2 = -\frac{1}{2}\sigma\sqrt{T} < 0$, so the ATM point lies strictly in the saddle region — $\Gamma > 0$ and Volga > 0 coexist with a negative eigenvalue produced by vanna.*

3 The at-the-money factorization

Set $m = 0$ (so $K = S$): then $d_1 = -d_2 = \frac{1}{2}\sigma\sqrt{T}$ and $d_1 d_2 = -\frac{1}{4}\sigma^2 T$. Writing $\beta := S\sigma T$,

$$H = \varphi\sqrt{T}\, M, \qquad M = \begin{pmatrix} \dfrac{1}{\beta} & \dfrac{1}{2} \\ \dfrac{1}{2} & -\dfrac{\beta}{4} \end{pmatrix}, \qquad \det M = -\frac{1}{\beta}\cdot\frac{\beta}{4} - \frac{1}{4} = -\frac{1}{2}.$$

Proposition 2. $\det M = -\frac{1}{2}$ *for every* $\beta > 0$. *With* $t := \operatorname{tr} M = \dfrac{1}{\beta} - \dfrac{\beta}{4}$, *the eigenvalues and axes of M are*

$$\lambda_\pm = \frac{t \pm \sqrt{t^2 + 2}}{2}, \qquad \lambda_+\lambda_- = -\tfrac{1}{2}, \qquad v_\pm \propto \Big(1,\ 2(\lambda_\pm - \tfrac{1}{\beta})\Big), \qquad \tan 2\theta = \frac{4\beta}{4 + \beta^2},$$

where θ is the rotation of the eigenframe from the (S, σ) axes.

Proof. $\lambda_\pm$ solve $\lambda^2 - t\lambda + \det M = 0$ with $\det M = -\frac{1}{2}$, so the discriminant is $t^2 - 4\det M = t^2 + 2$ and $\lambda_+\lambda_- = \det M = -\frac{1}{2}$. For the axis, $(M - \lambda I)v = 0$ gives $v \propto (M_{12},\ \lambda - M_{11}) = (\frac{1}{2}, \lambda - \frac{1}{\beta})$, i.e. $(1, 2(\lambda - \frac{1}{\beta}))$; and $\tan 2\theta = 2M_{12}/(M_{11} - M_{22}) = 4\beta/(4 + \beta^2)$ for the symmetric 2×2 case. □

The determinant invariant $\det M = -\frac{1}{2}$ means the two axes carry eigenvalues of strictly opposite sign for every β: the ATM curvature is a saddle whose principal-axis product is fixed independent of spot, volatility, and maturity.

4 Coordinate dependence and the natural (log) normalization

H mixes units (Γ per S^2, Volga per σ^2, Vanna per $S\sigma$), so its eigenvectors are *not* invariant: a rescaling by $D = \operatorname{diag}(a, b)$ sends $H \mapsto D^{-1}HD^{-1}$, changing both axes and eigenvalues. By Sylvester's law of inertia only the signature (Proposition 1) survives. The natural choice measures curvature against *relative* moves dS/S, $d\sigma/\sigma$ — the congruence $\tilde{H} = DHD$, $D = \operatorname{diag}(S, \sigma)$, so $\tilde{H}_{11} = S^2\Gamma$, $\tilde{H}_{12} = S\sigma\,\text{Vanna}$, $\tilde{H}_{22} = \sigma^2\,\text{Volga}$.

Proposition 3. *At the money, with $w := \sigma^2 T$ the total variance,*

$$\tilde{H} = S\sigma\varphi\sqrt{T}\, M_w, \qquad M_w = \begin{pmatrix} \dfrac{1}{w} & \dfrac{1}{2} \\ \dfrac{1}{2} & -\dfrac{w}{4} \end{pmatrix}, \qquad \det M_w = -\tfrac{1}{2}, \qquad \tan 2\theta = \frac{4w}{4 + w^2}.$$

The prefactor is exactly $S\sigma\varphi\sqrt{T}$ — not *$\varphi\sqrt{T}$; the factor $S\sigma$ is the Jacobian of the log rescaling and must be carried.*

Proof. At $m = 0$: $\tilde{H}_{12} = S\sigma \cdot (-\varphi d_2/\sigma) = -S\varphi d_2 = \frac{1}{2}S\sigma\varphi\sqrt{T}$, fixing the prefactor $P = S\sigma\varphi\sqrt{T}$. Then $\tilde{H}_{11} = S^2\Gamma = S\varphi/(\sigma\sqrt{T}) = P/w$ and $\tilde{H}_{22} = \sigma^2\,\text{Volga} = -S\varphi\sigma^3 T^{3/2}/4 = -Pw/4$, so $\tilde{H} = P\,M_w$; the angle formula follows from Proposition 2 with $\beta \to w$. □

So the log-normalized ATM curvature has the identical M-form with β replaced by the total variance $w = \sigma^2 T$. For a small total variance $w \to 0$, using $\det M_w = \lambda_+\lambda_- = -\frac{1}{2}$:

$$\lambda_+ = \frac{1}{w} + \frac{w}{4} + O(w^3) \;\approx\; \frac{1}{w} \;\text{ (gamma axis)}, \qquad \lambda_- = -\frac{1}{2\lambda_+} \;\approx\; -\frac{w}{2} \;\text{ (volga axis)}, \qquad \theta \approx \frac{w}{2}.$$

The volga *entry* is $-w/4$, but vanna's off-diagonal contributes an equal $-w/4$ to the eigenvalue, so the volga-axis eigenvalue is $\lambda_- \approx -w/2$, twice the entry — the exact invariant $\lambda_+\lambda_- = -\frac{1}{2}$ forces it. Interpretation: at the money the gamma and volga directions nearly decouple ($\lambda_+\lambda_-$ fixed, axes nearly the coordinate axes), and vanna's entire effect is to rotate the eigenframe by $\theta \approx w/2$, half the total variance.

5 Numerical check

Take $S = K = 100$, $\sigma = 0.2$, $T = 1$, so $\beta = S\sigma T = 20$ and $\varphi\sqrt{T} = 0.396953$; closed form and `numpy.linalg.eigh` agree.

Quantity	Closed form	Numeric
$H_{11} = \Gamma$	$\varphi\sqrt{T}/\beta = 0.019848$	0.019848
$H_{12} = \text{Vanna}$	$\varphi\sqrt{T} \cdot \frac{1}{2} = 0.198476$	0.198476
$H_{22} = \text{Volga}$	$-\varphi\sqrt{T} \cdot \frac{\beta}{4} = -1.984763$	-1.984763
$\det H$	$(\varphi^2\sqrt{T}/\sigma)\, d_2 = -0.078786$	-0.078786
$\lambda_-,\ \lambda_+$	$\varphi\sqrt{T} \cdot \{\frac{t \pm \sqrt{t^2+2}}{2}\}$	-2.004225, 0.039310

Here $\det H < 0$ confirms the ATM saddle (Proposition 1, $d_2 = -0.1 < 0$), and $\lambda_-\lambda_+ = (\varphi\sqrt{T})^2 \det M = -\frac{1}{2}(\varphi\sqrt{T})^2$ recovers the invariant.